Mitigating the Effects of Rising Temperatures on Community Health

Extreme heat may pose a danger to human health and has been the direct cause of death for 11,000 Americans since 1979. In warm climates, the human body regulates temperature by employing mechanisms like sweating to dissipate heat. When the body is incapable of these heat-releasing mechanisms, it can overheat, leading to illness. Heat-related illness — broadly called hyperthermia—is a condition where the body cannot cool itself, leading to a rapid rising of internal body temperature.

The Federal Emergency Management Agency (FEMA) classifies extreme heat as a period of two or more days with high humidity and temperatures maintaining above 90℉. The Centers for Disease Control and Prevention (CDC) reports that extreme heat is the leading cause of weather-related death in the United States. Many public health officials suggest that these deaths are underreported. Hyperthermia can also exacerbate pre-existing respiratory and cardiac issues, worsening health outcomes. With 2021 marking the second warmest year ever recorded, heat-related health incidents increased.

Impacted Communities

Extreme heat-related illness risk in a community is determined by a combination of factors, including exposure to high temperatures, level of population acclimation to high heat, and availability of infrastructure to reduce temperature. Individuals living without proper ventilation or air conditioning, young children, older adults, athletes, outdoor laborers, people with pre-existing/chronic health conditions, and some communities of color are at higher risk in extreme heat. Affected districts report power supply strains, overwhelmed healthcare systems, and excess deaths.

Existing Programs to Address High Heat

The Occupational Safety and Health Administration (OSHA) lays out recommendations and guidelines for employers that require shade and water availability, as well as mitigation efforts that lessen the risk for laborers. The Department of Homeland Security (DHS), the Department of Health and Human Services (HHS), the Department of Housing and Urban Development (HUD), and the Department of Agriculture all provide avenues for federal assistance for communities facing extreme heat emergencies.

The Asuncion Valdivia Heat Illness and Fatality Prevention Act of 2021 created requirements at the Department of Labor for prevention and response training on heat-related illness, expanding federal recognition. The Preventing Heat Illness and Deaths Act of 2021 also supports preventative measures against heat-related illness by building capacity that mitigates the impact of extreme heat. The Act also provides funding for additional research and education programs, so more robust response and prevention programs can be created in the future.

FEMA's Hazard Mitigation Assistance programs (namely Public Assistance, the Hazard Mitigation Grant Program, and Building Resilient Infrastructure and Communities) can fund high-risk, heat impacted communities. Additionally, the President may authorize FEMA to provide additional assistance through the Stafford Act in emergencies, though it has never done so in response to extreme heat.

The CDC offers a variety of grants for preventative efforts, including the Public Health Emergency Preparedness (PHEP) Cooperative Agreement, the Climate-Ready Cities and States Initiative, and the Preventive Health and Health Services (PHHS) Block Grant, among others that provide funds directly to communities taking efforts to prevent health-related death and illness.

Preventative Programs and Initiatives

The Environmental Protection Agency (EPA) and the National Oceanic and Atmospheric Administration (NOAA) report a projected upward trend in temperature as climate change continues. In the past 50 years, the frequency of heatwaves has increased. Geographically, the North, the West, and Alaska have seen the most significant shifts in heat levels in the past two decades.

Investments in emissions-free technologies have continued in recent years, with Fixing America's Surface Transportation Act (FAST) being a notable contributor to reducing emissions from cars — a significant contributor to extreme heat, especially in urban areas. Replacing conventional asphalt pavements with less heat absorbent materials or painting the asphalt with less absorbent colors (such as green or blue) has also been shown to significantly reduce heat in urban settings in conjunction with investments into alternative energy vehicles. Planting trees in cities — both to absorb CO_2 emissions and provide shade — has also been connected to reducing hyperthermia cases in the past.

Future Response Initiatives

The CDC suggests that air conditioning and insulation may be the most effective preventative measure against hyperthermia. Solutions that leverage these air conditioning and insulation measures may save money in health costs and improve health and quality of living in affected communities. Expanding the scope of the Community Development Block Grant Program (CDBG) through HUD or creating additional grant programs to further cooling measures may aid in prevention efforts. However, installing air conditioning units may increase energy usage, contributing to climate concerns and possibly further exacerbating extreme heat risk in the future. Expanding other grants that only support new developments may not aid existing housing, possibly excluding many high-risk communities.

Electric car emissions create only 19.8% of the total heat emitted by conventional vehicles (gas-powered) per mile. In urban areas abroad (such as Beijing), replacing traditional cars with electric vehicles reduced urban heat considerably, reduced electricity consumed by air conditioners by 14.44 million kilowatt-hours, and reduced daily CO_2 emissions by 10,686 tons. However, these investments may be costly and have limited effectiveness in more rural communities where car emissions are not concentrated to high enough degrees to so heavily impact temperature or weather.

Increased capacity for building emergency response programs could benefit many by providing additional support to the most vulnerable and under-resourced communities. Still, it may be difficult to tailor to the needs of localities. Though extreme heat incidents are anticipated to increase, investments in prevention programs that reduce community heat or create greater protection from heat exposure may be more cost-effective and produce better results than programs that only respond to heat emergencies as they occur.

As current programs stand, capacity is continuing to be built to both prevent heat-related illness and respond to extreme heat health emergencies. With climate conditions continuing to vary, a continued review of the current federal and state initiatives may be beneficial.